OpenTelemetry Collector Architecture

Series: OTEL | **Notebook:** 2 of 8 | **Created:** January 2026 | **Last Updated:** 02/09/2026

Understanding the OTel Collector Pipeline

The OpenTelemetry Collector is the backbone of OTel deployments—a vendor-agnostic service for receiving, processing, and exporting telemetry data. This notebook covers its architecture, components, and configuration.

Table of Contents

Prerequisites

Requirement	Details
Knowledge	OTEL-01: OpenTelemetry Fundamentals
Tools	YAML familiarity for configuration

1. Collector Overview

What is the Collector?

The OpenTelemetry Collector is a standalone service that:

Function	Description
Receives	Ingests telemetry from multiple sources
Processes	Transforms, filters, enriches data
Exports	Sends to one or more backends

Collector Distributions

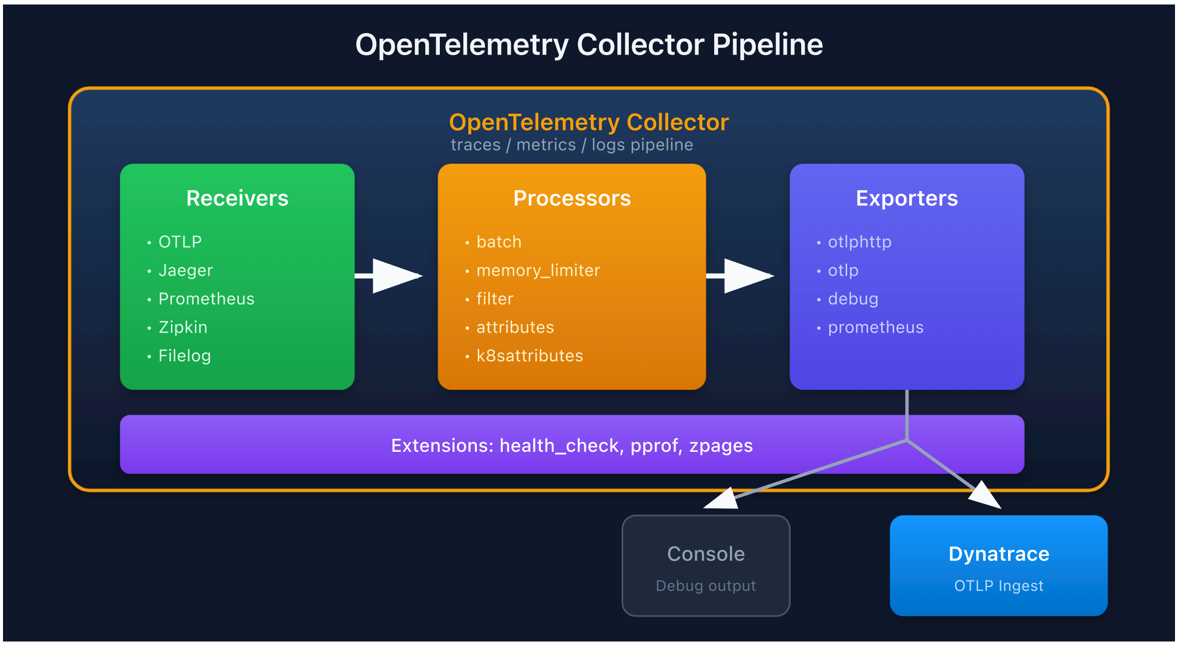
Distribution	Use Case	Components
Core	Minimal, essential only	Basic receivers/exporters
Contrib	Extended functionality	Many community components
Custom	Your specific needs	Build with ocb (OTel Collector Builder)
Vendor	Pre-configured for vendor	Dynatrace, AWS, etc.

Why Use a Collector?

Benefit	Description
Decoupling	Apps don't need backend-specific SDKs
Processing	Centralized filtering, sampling, enrichment
Fan-out	Send to multiple backends
Buffering	Retry and batching for reliability
Security	Centralize credentials and auth

2. Pipeline Architecture

Data Flow



Pipeline Types

Pipeline	Data Type	Example
`traces`	Distributed traces	Request spans
`metrics`	Quantitative measurements	CPU usage
`logs`	Log records	Application logs

Each signal type has its own pipeline(s) with separate receivers, processors, and exporters.

3. Receivers

Receivers ingest telemetry data from external sources.

Common Receivers

Receiver	Protocol	Signals	Use Case
`otlp`	OTLP (gRPC/HTTP)	Traces, Metrics, Logs	OTel SDKs
`jaeger`	Jaeger (gRPC/HTTP)	Traces	Jaeger clients
`zipkin`	Zipkin HTTP	Traces	Zipkin clients
`prometheus`	Prometheus scrape	Metrics	Prometheus targets
`filelog`	File tailing	Logs	Log files
`hostmetrics`	Host stats	Metrics	CPU, memory, disk

OTLP Receiver Configuration

```
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
      http:
        endpoint: 0.0.0.0:4318
```

Prometheus Receiver

```
receivers:
  prometheus:
    config:
      scrape_configs:
        - job_name: 'my-app'
          scrape_interval: 15s
          static_configs:
            - targets: ['app:8080']
```

4. Processors

Processors transform, filter, or enrich telemetry data.

Essential Processors

Processor	Purpose	When to Use
`batch`	Batch data for efficiency	Always
`memory_limiter`	Prevent OOM	Always
`resourcedetection`	Add resource attributes	K8s, cloud
`attributes`	Add/modify attributes	Enrichment
`filter`	Drop unwanted data	Cost reduction
`transform`	OTTL-based transformations	Complex processing

Batch Processor (Critical)

```
processors:
  batch:
    timeout: 10s
    send_batch_size: 1000
    send_batch_max_size: 1500
```

Memory Limiter (Critical)

```
processors:
  memory_limiter:
    check_interval: 1s
    limit_mib: 400
    spike_limit_mib: 100
```

Attributes Processor

```
processors:
  attributes:
    actions:
      - key: environment
        value: production
        action: upsert
      - key: api.key
        action: delete
```

Filter Processor

```
processors:
  filter:
    traces:
      span:
        - 'attributes["http.url"] == "/health"'
```

5. Exporters

Exporters send processed telemetry to backends.

Common Exporters

Exporter	Target	Signals	Notes
`otlphttp`	OTLP over HTTP	All	Dynatrace, many backends
`otlp`	OTLP over gRPC	All	Lower overhead
`debug`	Console/logs	All	Development
`prometheus`	Prometheus endpoint	Metrics	Pull model
`file`	Local files	All	Debugging

OTLP HTTP Exporter (Dynatrace)

```
exporters:
  otlphttp:
    endpoint: https://{your-env}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_API_TOKEN}
```

Debug Exporter

```
exporters:
  debug:
    verbosity: detailed
    sampling_initial: 5
    sampling_thereafter: 200
```

Multiple Exporters (Fan-Out)

```
exporters:
  otlphttp/dynatrace:
    endpoint: https://${DT_ENV_ID}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_API_TOKEN}
  otlphttp/backup:
    endpoint: https://backup-backend.example.com

service:
  pipelines:
    traces:
      receivers: [otlp]
      processors: [batch]
      exporters: [otlphttp/dynatrace, otlphttp/backup]  # Both!
```

6. Extensions

Extensions provide operational capabilities for the Collector itself.

Common Extensions

Extension	Purpose
`health_check`	Liveness/readiness endpoints
`pprof`	Go profiling data
`zpages`	Debug pages for pipelines

> **Note:** The `memory_ballast` extension is **deprecated** and was removed from default Helm chart configurations. Use the `GOMEMLIMIT` environment variable instead (set to ~80% of the container memory limit). See Collector discussion #9264.

Configuration

```
extensions:
  health_check:
    endpoint: 0.0.0.0:13133
  pprof:
    endpoint: 0.0.0.0:1777
  zpages:
    endpoint: 0.0.0.0:55679

service:
  extensions: [health_check, pprof, zpages]
```

7. Connectors

Connectors link pipelines, acting as both exporter and receiver.

Use Cases

Connector	Function
`spanmetrics`	Generate metrics from spans
`servicegraph`	Build service dependency graph
`count`	Count data points

Span Metrics Connector

```yaml
connectors:
  spanmetrics:
    dimensions:
      - name: http.method
      - name: http.status_code
    histogram:
      explicit:
        buckets: [100ms, 500ms, 1s, 5s]

service:
  pipelines:
    traces:
      receivers: [otlp]
      exporters: [spanmetrics, otlphttp]
    metrics:
      receivers: [spanmetrics]  # Receives from connector
      exporters: [otlphttp]
```

8. Configuration Basics

Minimal Configuration

```yaml
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
      http:
        endpoint: 0.0.0.0:4318

processors:
  batch:
    timeout: 10s

exporters:
  otlphttp:
    endpoint: https://${DT_ENV_ID}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_API_TOKEN}

service:
  pipelines:
    traces:
      receivers: [otlp]
      processors: [batch]
      exporters: [otlphttp]
    metrics:
      receivers: [otlp]
      processors: [batch]
      exporters: [otlphttp]
    logs:
      receivers: [otlp]
      processors: [batch]
      exporters: [otlphttp]
```

Configuration Best Practices

Practice	Reason
Always use `batch` processor	Efficiency
Always use `memory_limiter`	Stability
Put `memory_limiter` first	Fail fast
Use environment variables for secrets	Security
Validate config before deploy	Catch errors early

Summary

In this notebook, you learned:

- Collector overview and distributions
- Pipeline architecture: receivers → processors → exporters
- Common receivers (OTLP, Prometheus, Jaeger)
- Essential processors (batch, memory_limiter, filter)
- Exporters for Dynatrace and other backends
- Extensions for operational capabilities
- Connectors for pipeline linking
- Configuration fundamentals

Next Steps

Next Notebook	Topic
OTEL-03: Collector Deployment	Deployment patterns
OTEL-04: Trace Instrumentation	Instrumenting code

References

- OTel Collector Documentation
- Collector Configuration
- Collector Releases

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.